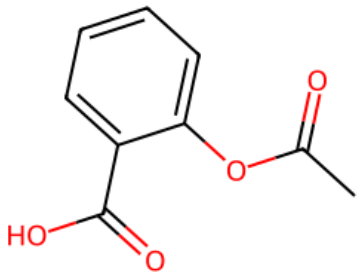
Understanding Molecular Feature Engineering

Morgan Fingerprints · Physicochemical Descriptors · Scaffold-Based Data Splitting

■ Running Example: Aspirin (CC(=O)Oc1ccccc1C(=O)O)

We use **Aspirin (Acetylsalicylic acid)** as a concrete example throughout this handout. Aspirin is a well-known pain reliever with MW = 180.16 g/mol. Its SMILES string encodes every atom and bond, and we convert this text into numbers that a machine-learning model can understand.



SMILES (Simplified Molecular Input Line Entry System) is a text notation that encodes a molecule's structure. RDKit parses this string into an internal graph of atoms and bonds — the starting point for all feature extraction.

Think of a molecule as a **graph**: atoms are *nodes*, bonds are *edges*. Every feature method below reads this same graph.

1. Morgan Fingerprints (Extended Connectivity Fingerprints — ECFP)

A Morgan fingerprint is a **fixed-length binary vector** (e.g. 2048 bits). Each bit represents whether a particular circular chemical sub-environment exists in the molecule. We use **radius = 2** (ECFP4), meaning each atom 'looks' at neighbours up to 2 bonds away.

Step	What Happens	Aspirin Example
Radius 0 (atom itself)	Each heavy atom gets a unique integer based on its own properties: atomic number, charge, mass, aromaticity.	C in –COOH gets hash 441
Radius 1 (1-bond away)	Each atom hashes itself together with its immediate neighbours to capture local bonding environment.	–COOH carbon + bonded O, OH
Radius 2 (2-bonds away)	Each atom hashes itself with neighbours up to 2 bonds away — captures ring membership, branching.	Benzene carbon sees the whole ring
Bit folding (final vector)	All collected hashes are mapped (modulo 2048) into a bit array. Bit = 1 if that sub-environment exists, else 0.	2048-element array, e.g. [...,0,1,0,1,...]

Why useful? Similar molecules → similar bit patterns → similar model predictions. RDKit API: `AllChem.GetMorganFingerprintAsBitVect(mol, radius=2, nBits=2048)`. Output shape: **(N molecules × 2048 bits)**.

2. Physicochemical Descriptors (9 human-interpretable numbers)

Descriptors are **computed physical/chemical properties**. Unlike fingerprints they are continuous numbers with clear scientific meaning — directly usable by toxicologists to explain a model's decision.

Descriptor	Aspirin Value	Definition	Toxicity Relevance
Molecular Weight	180.16 g/mol	Total mass of all atoms	Predicts membrane permeability
LogP	1.31	Water vs. octanol partition coeff.	Lipophilicity; cell penetration
TPSA	63.60 ■	Surface area of polar atoms (N,O,H)	Oral bioavailability / absorption
H-Bond Donors (HBD)	1	N–H or O–H groups that donate H	Solubility, protein binding
H-Bond Acceptors (HBA)	3	N or O atoms that accept H	Solubility, protein binding
Rotatable Bonds	2	Freely rotating non-ring bonds	Molecular flexibility

Descriptor	Aspirin Value	Definition	Toxicity Relevance
Ring Count	1	Number of rings (benzene = 1)	Rigidity, scaffold class
Heavy Atom Count	13	All non-hydrogen atoms	Molecular size
Fsp^3	0.11	Fraction of sp^3 carbons	3-D character; drug-likeness

Output shape: **(N molecules × 9 descriptors)**. RDKit functions: `Descriptors.MolWt`, `Descriptors.MolLogP`, `rdMolDescriptors.CalcTPSA`, …

3. Combined Feature Matrix (2048 + 9 = 2057 features)

We concatenate both representations horizontally. The fingerprint bits capture *structural patterns* (presence of sub-environments), while descriptors capture *global physicochemical properties*. Together they give the ML model complementary information.

Representation	Feature Dim.	When to Use
Fingerprints only	2 048	Structural sub-environments; captures shape & topology
Descriptors only	9	Human-interpretable chemistry; often preferred by toxicologists
Combined (concat.)	2 057	Best of both worlds — used as the primary feature set

4. Bemis-Murcko Scaffold Splitting (Why random splits are wrong)

In drug discovery the **real test** is whether a model works on chemically *new* molecules. A random 80/10/10 split often places very similar molecules in both train and test, inflating accuracy. Scaffold splitting groups molecules by their core ring system (*scaffold*) so train and test share **no scaffolds**.

Aspirin's Bemis-Murcko scaffold = benzene ring (c1ccccc1).
*All aspirin-like molecules share this scaffold and go to the **same** split.*

Step	Action
1. Extract scaffold	RDKit strips all side-chains, keeping only the ring system
2. Group molecules	All molecules with the same scaffold form one cluster
3. Sort clusters	Clusters sorted by size (largest first) for reproducibility
4. Assign splits	Train 80 % / Val 10 % / Test 10 % — NO scaffold in two groups
5. Seed = 42	Reproducible: same split every run; reported in paper

Result: A rigorous, publication-standard split that tests true generalisation. Implemented with `deepchem.splits.ScaffoldSplitter` + custom fallback using RDKit `MurckoScaffold.GetScaffoldForMol()`.